A First Course in Abstract Algebra

Section 9: Orbits, Cycles, and the Alternating Groups

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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9 Orbits, Cycles, and the Alternating Groups

Conventions. Permutations compose from right to left. The identity is e. A group generated by an element a is denoted $\langle a \rangle$.

Exercise 9.1: Orbits of a finite permutation

Plan. Start at an unused symbol, repeatedly apply the permutation until it returns, and repeat with a symbol not yet used. The nontrivial cycles are $(1, 5, 2)(4, 6)$. Each fixed point forms a singleton orbit, so the complete orbit list is

$$\{1, 2, 5\}, \quad \{4, 6\}, \quad \{3\}.$$

Every symbol occurs in exactly one listed set; negative powers move backwards within the same set and produce no further points.

Exercise 9.2: Orbits of a finite permutation

Plan. Start at an unused symbol, repeatedly apply the permutation until it returns, and repeat with a symbol not yet used. The nontrivial cycles are $(1, 5, 8, 7)(2, 6, 3)$. Each fixed point forms a singleton orbit, so the complete orbit list is

$$\{1, 5, 7, 8\}, \quad \{2, 3, 6\}, \quad \{4\}.$$

Every symbol occurs in exactly one listed set; negative powers move backwards within the same set and produce no further points.

Exercise 9.3: Orbits of a finite permutation

Plan. Start at an unused symbol, repeatedly apply the permutation until it returns, and repeat with a symbol not yet used. The nontrivial cycles are $(1, 2, 3, 5, 4)(7, 8)$. Each fixed point forms a singleton orbit, so the complete orbit list is

$$\{1, 2, 3, 4, 5\}, \quad \{7, 8\}, \quad \{6\}.$$

Every symbol occurs in exactly one listed set; negative powers move backwards within the same set and produce no further points.

Exercise 9.4: Orbits of the integer shift by 1

The rth power sends an integer a to $a + 1r$, where r ranges over $\mathbb{Z}$. Hence the orbit of a is $a + 1\mathbb{Z}$, its residue class modulo 1. The distinct orbits are

$$\mathbb{Z}.$$

The sign of the shift only changes the direction in which these same points are traversed.

Exercise 9.5: Orbits of the integer shift by 2

The rth power sends an integer a to $a + 2r$, where r ranges over $\mathbb{Z}$. Hence the orbit of a is $a + 2\mathbb{Z}$, its residue class modulo 2. The distinct orbits are

$$0 + 2\mathbb{Z}, \quad 1 + 2\mathbb{Z}.$$

The sign of the shift only changes the direction in which these same points are traversed.

Exercise 9.6: Orbits of the integer shift by -3

The rth power sends an integer a to $a + -3r$, where r ranges over $\mathbb{Z}$. Hence the orbit of a is $a + 3\mathbb{Z}$, its residue class modulo 3. The distinct orbits are

$$0 + 3\mathbb{Z}, \quad 1 + 3\mathbb{Z}, \quad 2 + 3\mathbb{Z}.$$

The sign of the shift only changes the direction in which these same points are traversed.

Exercise 9.7: A product of cycles

Plan. Read the factors from right to left, leaving a symbol unchanged whenever it is absent from a factor. The images of the eight arguments are

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 1 & 3 & 5 & 8 & 6 & 2 & 7 \end{pmatrix}.$$

Following each chain of these images yields the disjoint-cycle answer

$$(1, 4, 5)(7, 8)(2, 5, 7) = (1, 4, 5, 8, 7, 2).$$

Any symbol omitted from the final cycles is fixed. The two-row form provides a check of every image, not just of one cycle.

Exercise 9.8: A product of cycles

Plan. Read the factors from right to left, leaving a symbol unchanged whenever it is absent from a factor. The images of the eight arguments are

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 7 & 2 & 8 & 5 & 4 & 1 & 6 \end{pmatrix}.$$

Following each chain of these images yields the disjoint-cycle answer

$$(1, 3, 2, 7)(4, 8, 6) = (1, 3, 2, 7)(4, 8, 6).$$

Any symbol omitted from the final cycles is fixed. The two-row form provides a check of every image, not just of one cycle.

Exercise 9.9: A product of cycles

Plan. Read the factors from right to left, leaving a symbol unchanged whenever it is absent from a factor. The images of the eight arguments are

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 4 & 3 & 7 & 8 & 6 & 2 & 1 \end{pmatrix}.$$

Following each chain of these images yields the disjoint-cycle answer

$$(1,2)(4,7,8)(2,1)(7,2,8,1,5) = (1,5,8)(2,4,7).$$

Any symbol omitted from the final cycles is fixed. The two-row form provides a check of every image, not just of one cycle.

Exercise 9.10: Disjoint cycles and transpositions

1. Follow the images.

The nontrivial chains are

$$1 \mapsto 8 \mapsto 1; \quad 3 \mapsto 6 \mapsto 4 \mapsto 3; \quad 5 \mapsto 7 \mapsto 5.$$

Thus the disjoint-cycle expression is $(1,8)(3,6,4)(5,7)$; any missing point is fixed.

2. Split each cycle.

Use $(a_1, a_2, \ldots, a_k) = (a_1, a_k) \cdots (a_1, a_3)(a_1, a_2)$, verified by following each a_i. This gives

$$(1,8)(3,6,4)(5,7) = (1,8)(3,4)(3,6)(5,7).$$

Exercise 9.11: Disjoint cycles and transpositions

1. Follow the images.

The nontrivial chains are

$$1 \mapsto 3 \mapsto 4 \mapsto 1; \quad 2 \mapsto 6 \mapsto 2; \quad 5 \mapsto 8 \mapsto 7 \mapsto 5.$$

Thus the disjoint-cycle expression is $(1,3,4)(2,6)(5,8,7)$; any missing point is fixed.

2. Split each cycle.

Use $(a_1, a_2, \ldots, a_k) = (a_1, a_k) \cdots (a_1, a_3)(a_1, a_2)$, verified by following each a_i. This gives

$$(1,3,4)(2,6)(5,8,7) = (1,4)(1,3)(2,6)(5,7)(5,8).$$

Exercise 9.12: Disjoint cycles and transpositions

1. Follow the images.

The nontrivial chains are

$$1 \mapsto 3 \mapsto 4 \mapsto 7 \mapsto 8 \mapsto 6 \mapsto 5 \mapsto 2 \mapsto 1.$$

Thus the disjoint-cycle expression is $(1, 3, 4, 7, 8, 6, 5, 2)$; any missing point is fixed.

2. Split each cycle.

Use $(a_1, a_2, \ldots, a_k) = (a_1, a_k) \cdots (a_1, a_3)(a_1, a_2)$, verified by following each a_i. This gives

$$(1, 3, 4, 7, 8, 6, 5, 2) = (1, 2)(1, 5)(1, 6)(1, 8)(1, 7)(1, 4)(1, 3).$$

Exercise 9.13: Orders of cycles and permutations

(a) $(1, 4, 5, 7)$ has order 4: four applications return each listed point, and no smaller positive number returns 1.

(b) A cycle of length k has order k. Each application advances one step, so a power fixes the cycle exactly when the exponent is divisible by k.

(c) The disjoint cycles of $\sigma = (4, 5)(2, 3, 7)$ have lengths $2, 3$, giving order 6. Those of $\tau = (1, 4)(3, 5, 7, 8)$ have lengths $2, 4$, giving order 4.

(d) Exercises 10, 11, 12 have nontrivial cycle lengths $(2, 3, 2)$, $(3, 2, 3)$, and (8), respectively. Their orders are $6, 6, 8$.

(e) The order of a permutation is the least common multiple of its disjoint-cycle lengths: a power is the identity precisely when it is a multiple of every cycle length.

Exercise 9.14: Maximum element order in S_5

Plan. Disjoint-cycle lengths are positive integers summing to 5, and the order is their least common multiple. Thus check partitions, rather than individual permutations. Grouping the partitions by their largest part L, the largest possible least common multiple in each group is

L	1	2	3	4	5
maximum	1	2	6	4	5

For example, these values can be reproduced by listing the remaining nondecreasing parts, each at most L, with sum $5 - L$, and taking the least common multiple with L. The overall maximum is 6, attained by disjoint cycles of lengths $2, 3$. This gives both an example and an exhaustive upper bound.

Exercise 9.15: Maximum element order in S_6

Plan. Disjoint-cycle lengths are positive integers summing to 6, and the order is their least common multiple. Thus check partitions, rather than individual permutations. Grouping the partitions by

their largest part L, the largest possible least common multiple in each group is

L	1	2	3	4	5	6
maximum	1	2	6	4	5	6

For example, these values can be reproduced by listing the remaining nondecreasing parts, each at most L, with sum $6 - L$, and taking the least common multiple with L. The overall maximum is 6, attained by disjoint cycles of lengths $1, 2, 3$. This gives both an example and an exhaustive upper bound.

Exercise 9.16: Maximum element order in S_7

Plan. Disjoint-cycle lengths are positive integers summing to 7, and the order is their least common multiple. Thus check partitions, rather than individual permutations. Grouping the partitions by their largest part L, the largest possible least common multiple in each group is

L	1	2	3	4	5	6	7
maximum	1	2	6	12	10	6	7

For example, these values can be reproduced by listing the remaining nondecreasing parts, each at most L, with sum $7 - L$, and taking the least common multiple with L. The overall maximum is 12, attained by disjoint cycles of lengths $3, 4$. This gives both an example and an exhaustive upper bound.

Exercise 9.17: Maximum element order in S_{10}

Plan. Disjoint-cycle lengths are positive integers summing to 10, and the order is their least common multiple. Thus check partitions, rather than individual permutations. Grouping the partitions by their largest part L, the largest possible least common multiple in each group is

L	1	2	3	4	5	6	7	8	9	10
maximum	1	2	6	12	30	12	21	8	9	10

For example, these values can be reproduced by listing the remaining nondecreasing parts, each at most L, with sum $10 - L$, and taking the least common multiple with L. The overall maximum is 30, attained by disjoint cycles of lengths $2, 3, 5$. This gives both an example and an exhaustive upper bound.

Exercise 9.18: Maximum element order in S_{15}

Plan. Disjoint-cycle lengths are positive integers summing to 15, and the order is their least common multiple. Thus check partitions, rather than individual permutations. Grouping the partitions by their largest part L, the largest possible least common multiple in each group is

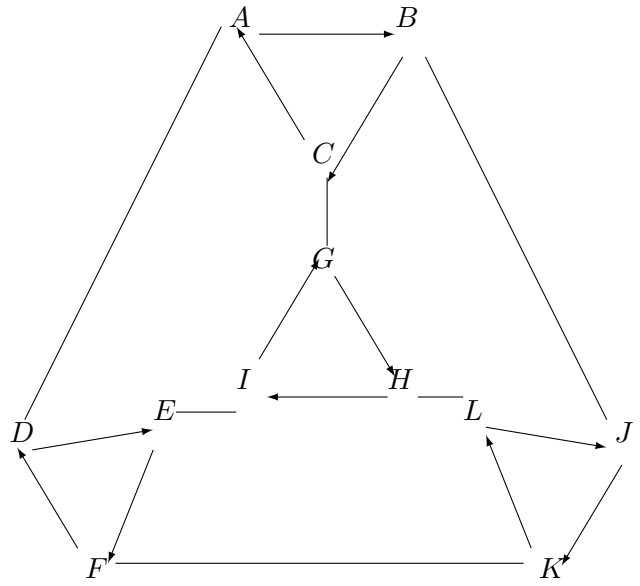
L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
maximum	1	2	6	12	60	60	105	56	45	30	44	12	26	14	15

For example, these values can be reproduced by listing the remaining nondecreasing parts, each at most L, with sum $15 - L$, and taking the least common multiple with L. The overall maximum is 105, attained by disjoint cycles of lengths $3, 5, 7$. This gives both an example and an exhaustive upper bound.

Exercise 9.19: Labeling the Cayley digraph of A_4

Put $a = (1, 2, 3)$ and $b = (1, 2)(3, 4)$. Follow each solid arrow by right multiplying by a, and each dashed edge by right multiplying by b. The following table labels every position in the supplied diagram; the letters also appear in the schematic below.

vertex	position	permutation
A	$topleft$	e
B	$topright$	$(1, 2, 3)$
C	$belowtoppair$	$(1, 3, 2)$
D	$leftoutside$	$(1, 2)(3, 4)$
E	$leftinner$	$(2, 4, 3)$
F	$bottomleft$	$(1, 4, 3)$
G	$middletop$	$(2, 3, 4)$
H	$middleright$	$(1, 3)(2, 4)$
I	$middleleft$	$(1, 4, 2)$
J	$rightoutside$	$(1, 3, 4)$
K	$bottomright$	$(1, 2, 4)$
L	$rightinner$	$(1, 4)(2, 3)$



The six edges without arrowheads represent the dashed b edges; both directions give the same operation since $b^2 = e$. For instance $D = b$, $E = ba$, and $F = ba^2$. The twelve entries are distinct and exhaust A_4, checking that no vertex has been missed.

Exercise 9.20: Definition of an orbit

The proposed definition is correct when "minimal" means minimal under inclusion among nonempty invariant subsets: an orbit is a nonempty subset $O \subseteq A$ with $\sigma[O] = O$ and no proper nonempty

subset having that property. Indeed, $O_a = \{\sigma^k(a) : k \in \mathbb{Z}\}$ is invariant, and any nonempty invariant subset containing a point of O_a contains its entire two-sided orbit. Conversely, pick a in a minimal such set; its orbit lies inside the set and hence must equal it.

Exercise 9.21: Definition of a cycle

A cycle in S_n is a permutation with *at most one orbit having more than one element*. Its length is the size of its largest orbit. Fixed points outside the moving cycle produce additional singleton orbits, so "only one orbit" is too restrictive. The book includes the identity as a cycle of length 1.

Exercise 9.22: Definition of the alternating group

Specify the common finite underlying set: the alternating group A_n is the subgroup of S_n consisting of all even permutations of $\{1, \ldots, n\}$. Without specifying this set or n, "all even permutations" does not describe a single group. Closure follows because products and inverses of even permutations are even.

Exercise 9.23: True or false

(a) False: $(1, 2)(3, 4)$ has two nontrivial orbits.

(b) True: a cycle is a special kind of permutation.

(c) False: first one must prove that the parity of the number of transpositions is independent of the chosen expression; otherwise the definition could assign both parities to one permutation.

(d) False: $\langle(1, 2, 3, 4)\rangle$ has an odd permutation but its elements are e, a 4-cycle, a double transposition, and the inverse 4-cycle; none is a transposition.

(e) False: $|A_5| = 5!/2 = 60$.

(f) False: $S_2 = \{e, (1, 2)\}$ is cyclic.

(g) True: A_3 is cyclic of order 3.

(h) True: extend each permutation of $\{1, \ldots, 7\}$ by fixing 8.

(i) True: relabel the seven symbols other than 5, and extend by fixing 5; the particular omitted symbol is immaterial.

(j) False: the identity is even and so is absent, and the product of two odd permutations is even.

Exercise 9.24: The table of A_3

The even permutations are $\rho_0 = e$, $\rho_1 = (1, 2, 3)$, and $\rho_2 = (1, 3, 2)$. Each 3-cycle is a product of two transpositions, while μ_1, μ_2, μ_3 are single transpositions and are odd. Thus

	ρ_0	ρ_1	ρ_2
ρ_0	ρ_0	ρ_1	ρ_2
ρ_1	ρ_1	ρ_2	ρ_0
ρ_2	ρ_2	ρ_0	ρ_1

The entries follow from $\rho_1^3 = e$ and $\rho_2 = \rho_1^2$.

Exercise 9.25: Synopsis of the determinant proof

A permutation matrix obtained from the identity by k row transpositions has determinant $(-1)^k$, so two transposition expressions for the same permutation must have the same parity because one matrix cannot have determinants 1 and -1 simultaneously.

Exercise 9.26: Synopsis of the orbit-counting proof

Multiplying a permutation by one transposition merges two distinct orbits or splits one orbit in two, changing its number of orbits by exactly 1. Starting from the identity's n orbits, the parity of the number of transpositions is therefore determined by the final number of orbits and cannot depend on the expression.

Exercise 9.27: Bounds on numbers of transpositions

Let the disjoint-cycle lengths, including fixed points, be $k_1, \ldots, k_t$, with sum n. A k_i-cycle uses $k_i - 1$ transpositions, so the entire permutation uses

$$(k_1 - 1) + \cdots + (k_t - 1) = n - t.$$

(a) Since $t \geq 1$, this is at most $n - 1$. The identity can be represented by zero transpositions, the empty product. **(b)** A permutation which is not a cycle has at least two nontrivial orbits, hence $t \geq 2$, giving at most $n - 2$ transpositions. **(c)** If the permutation is odd, the displayed number $m = n - t$ is odd. Since $m \leq n - 1 < 2n + 3$ and $2n + 3 - m$ is even, insert $(2n + 3 - m)/2$ pairs $(1, 2)(1, 2)$ into that expression to obtain exactly $2n + 3$ factors. For an even permutation m is even and the same insertion of $(2n + 8 - m)/2$ pairs gives exactly $2n + 8$ factors. Each pair is the identity and so does not change the permutation.

Exercise 9.28: Merging orbits when a point is fixed

(a) Write the orbit containing j as $(j, x_1, \ldots, x_k)$ and suppose i is a different fixed point. Then direct calculation gives

$$(i, j)(j, x_1, \ldots, x_k) = (i, j, x_1, \ldots, x_k).$$

Thus the two old orbit diagrams and the new diagram can be drawn as

$$\underbrace{i \longrightarrow i}_{\text{one orbit}} \qquad \underbrace{j \longrightarrow x_1 \longrightarrow \cdots \longrightarrow x_k \longrightarrow j}_{\text{one orbit}} \quad \rightsquigarrow \quad i \longrightarrow j \longrightarrow x_1 \longrightarrow \cdots \longrightarrow x_k \longrightarrow i.$$

The other orbits are unchanged, so the orbit count decreases by one. **(b)** If j is fixed too, the two singleton loops $i \to i$ and $j \to j$ become the single 2-cycle $i \to j \to i$ under $(i, j)\sigma$. Again exactly two orbits have become one.

Exercise 9.29: Even permutations in a subgroup

If H has no odd permutations, every element is even. Otherwise fix an odd $\tau \in H$. Left multiplication by τ maps the even elements of H to odd elements of H, because signs multiply. Its inverse is left multiplication by τ^{-1}, which is also odd. Thus it is a bijection between the even and odd elements of H. As H is finite, exactly half its elements are even.

Exercise 9.30: How many points a cycle moves

A cycle of length $n > 1$ moves exactly its n listed points: each goes to the next distinct listed point, and all other points are fixed. Under the book's convention, a cycle of length 1 is the identity and moves 0 points. Thus the usual answer n requires the nontrivial-cycle condition $n > 1$.

Exercise 9.31: Finitely supported permutations form a subgroup

For a permutation σ, write $\operatorname{supp}(\sigma) = \{a : \sigma(a) \neq a\}$. The identity has empty support. If both σ and τ fix a point a, their product fixes a, so

$$\operatorname{supp}(\sigma\tau) \subseteq \operatorname{supp}(\sigma) \cup \operatorname{supp}(\tau).$$

A union of two finite sets is finite, proving product closure. A point is fixed by σ if and only if it is fixed by σ^{-1}, so the two supports are equal. Hence inverses also have finite support, and the given set is a subgroup.

Exercise 9.32: A uniform bound on support size

No. Choose 100 distinct points of the infinite set A, and let σ be a 50-cycle on the first 50, while τ is a 50-cycle on the other 50. Extend both by the identity elsewhere. Each moves exactly 50 points, so both belong to K. Their disjoint supports mean $\sigma\tau$ moves all 100 selected points, so it is not in K. Thus product closure fails.

Exercise 9.33: The odd permutations form a coset

Let σ be the fixed odd permutation and let π be any odd permutation. Set $\alpha = \sigma^{-1}\pi$. Both factors are odd, so their product is even and $\alpha \in A_n$. Then $\pi = \sigma\alpha$. Conversely, the product of σ with any even permutation is odd. Thus the set of odd permutations is exactly σA_n.

Exercise 9.34: Squaring an odd-length cycle

Let $\sigma = (a_0, a_1, \ldots, a_{n-1})$ with n odd. Its square moves a_i to $a_{i+2 \bmod n}$. If $2k \equiv 0 \pmod{n}$, oddness implies $n \mid k$, so the first positive return occurs after n steps. Hence all n symbols lie in one cycle of σ^2, and every point outside the original cycle stays fixed. For $n = 1$, both permutations are the identity, also a cycle by convention.

Exercise 9.35: When a power of a cycle is a cycle

Plan. Advancing r steps around an n-cycle partitions its n points into $d = \gcd(n, r)$ cycles, each of length n/d. Indeed, $\sigma^{rk}(a) = a$ precisely when $n \mid rk$, whose least positive solution is $k = n/d$. If $n/d > 1$, all d orbits are nontrivial, so the power is a cycle exactly when $d = 1$. There is an important convention in this text: the identity is also a cycle. If $n \mid r$, then $\sigma^r = e$ and it is therefore a cycle as well. With that convention the exact condition is

$$\boxed{\gcd(n, r) = 1 \quad \text{or} \quad n \mid r.}$$

If "cycle" is intended to mean a nonidentity n-cycle, the familiar condition is just $\gcd(n, r) = 1$ (for $n > 1$). The extra identity case must not be silently omitted under Definition 9.6.

Exercise 9.36: Left multiplication is a permutation

The inverse of $\lambda_a : g \mapsto ag$ is $\lambda_{a^{-1}}$, because $a^{-1}(ag) = g$ and $a(a^{-1}g) = g$. Equivalently, cancellation gives injectivity, and a desired target h is reached by the input $a^{-1}h$, giving surjectivity. Thus $\lambda_a \in S_G$.

Exercise 9.37: The subgroup of left translations

By the preceding exercise every λ_a is a permutation. The identity is λ_e. Associativity in G gives

$$(\lambda_a \lambda_b)(g) = a(bg) = (ab)g = \lambda_{ab}(g),$$

so the set is closed under composition. Finally $\lambda_a^{-1} = \lambda_{a^{-1}}$ also lies in it. These facts prove it is a subgroup of S_G.

Exercise 9.38: Left translations act transitively

Given $x, y \in G$, solve $ax = y$ by taking $a = yx^{-1}$. The left translation λ_a then sends x to y. Since this works for every pair of points in the set G, the left-translation subgroup is transitive on G.

Exercise 9.39: Two generators for the symmetric group

Put $c = (1, 2, \ldots, n)$ and $t = (1, 2)$. Conjugating a transposition relabels its entries, so

$$c^k t c^{-k} = (c^k(1), c^k(2)).$$

For $k = 0, \ldots, n-2$, these include all adjacent transpositions $s_i = (i, i+1)$, $1 \le i < n$. For $i < j$ one has

$$(i, j) = s_i s_{i+1} \cdots s_{j-2} s_{j-1} s_{j-2} \cdots s_{i+1} s_i.$$

To verify it, the product before the middle transposition conjugates $(j-1, j)$ to (i, j); alternatively follow i, j and each intermediate point. Thus every transposition lies in $\langle c, t \rangle$. Every permutation is a product of transpositions, so $\langle c, t \rangle = S_n$.